A-Level Mathematics — Pure 3

Key Function Graphs Reference Sheet

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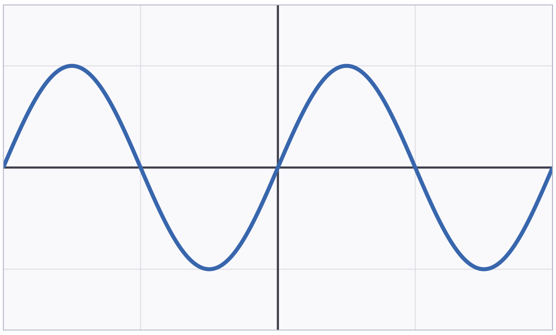
Covers all major graphs for A-Level Maths P3: sin x, cos x, tan x, cosec x, sec x, cot x, e^x and ln x. Red dashed lines indicate vertical asymptotes. All angles in radians.

Core Trigonometric Functions

y = sin x

Domain: All real numbers **Range:** [-1, 1]



- Period: 2π
- Amplitude: 1
- Odd function (rotational symmetry about origin)
- sin(0) = 0

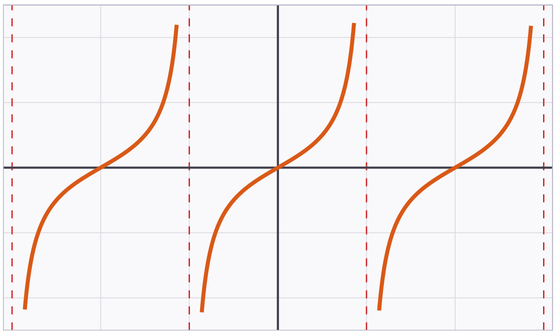
y = cos x

Domain: All real numbers **Range:** [-1, 1]

- Period: 2π
- Amplitude: 1
- Even function (symmetric about y-axis)
- cos(0) = 1

y = tan x

Domain: x ≠ π/2 + nπ (n ∈ Z) **Range:** All real numbers

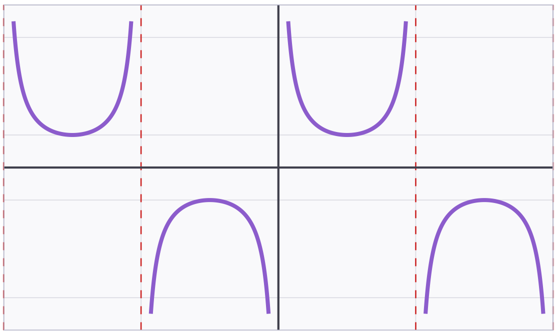


- Period: π
- Odd function
- Vertical asymptotes at x = ±π/2, ±3π/2
- Passes through origin

Reciprocal Trigonometric Functions

y = cosec x = 1/sin x

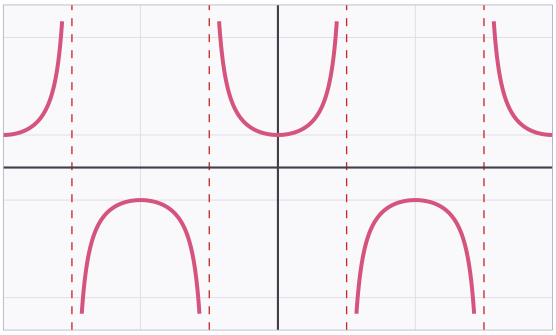
Domain: $x \neq n\pi$ ($n \in Z$) **Range:** $y \leq -1$ or $y \geq 1$



- Period: 2π
- Odd function
- Asymptotes at $x = 0, \pm\pi, \pm 2\pi$
- No values exist in the open interval (-1, 1)

y = sec x = 1/cos x

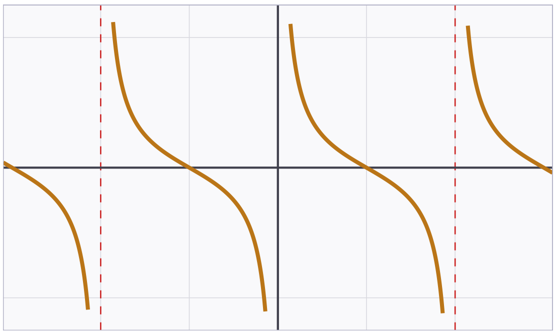
Domain: $x \neq \pi/2 + n\pi$ ($n \in Z$) **Range:** $y \leq -1$ or $y \geq 1$



- Period: 2π
- Even function (symmetric about y-axis)
- Asymptotes at $x = \pm\pi/2, \pm 3\pi/2$
- No values exist in the open interval (-1, 1)

y = cot x = cos x / sin x

Domain: $x \neq n\pi$ ($n \in Z$) **Range:** All real numbers

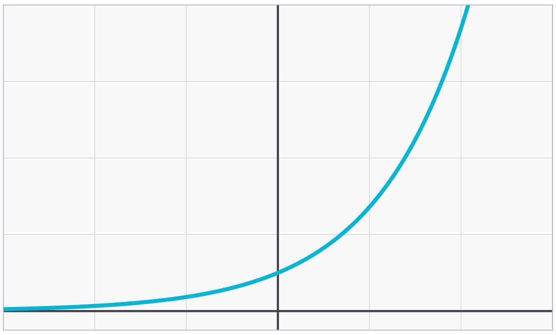


- Period: π
- Odd function
- Asymptotes at $x = 0, \pm\pi, \pm 2\pi$
- Decreasing on each interval between asymptotes

Exponential & Logarithmic Functions

y = e^x

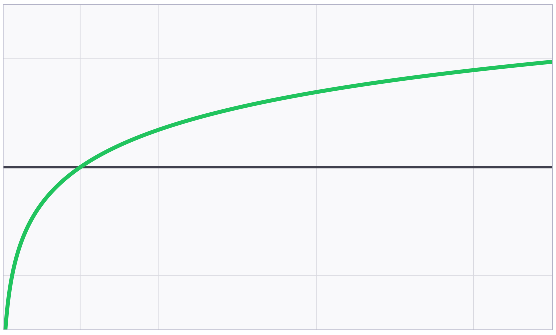
Domain: All real numbers **Range:** y > 0



- y-intercept: 1
- Horizontal asymptote: y = 0 as x → -∞
- Always increasing; never touches x-axis
- Inverse: y = ln x

y = ln x

Domain: x > 0 **Range:** All real numbers



- x-intercept: 1
- Vertical asymptote: x = 0
- Inverse of y = e^x (reflection in y = x)
- ln(e) = 1

Quick Reference Summary Table

Function	Period	Domain	Range	Symmetry
sin x	2π	All reals	$[-1, 1]$	Odd
cos x	2π	All reals	$[-1, 1]$	Even
tan x	π	$x \neq \pi/2 + n\pi$	All reals	Odd
cosec x	2π	$x \neq n\pi$	$y \leq -1$ or $y \geq 1$	Odd
sec x	2π	$x \neq \pi/2 + n\pi$	$y \leq -1$ or $y \geq 1$	Even
cot x	π	$x \neq n\pi$	All reals	Odd
e^x	—	All reals	$y > 0$	Neither
ln x	—	$x > 0$	All reals	Neither

Note: $n \in$ ■ denotes any integer. All angles are in radians.